FIREWALL, NAT AND PACKET FILTERING

Network Security – Practical Assignment

TASK 1

Set up Windows Defender Firewall on a Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule created.

Answer

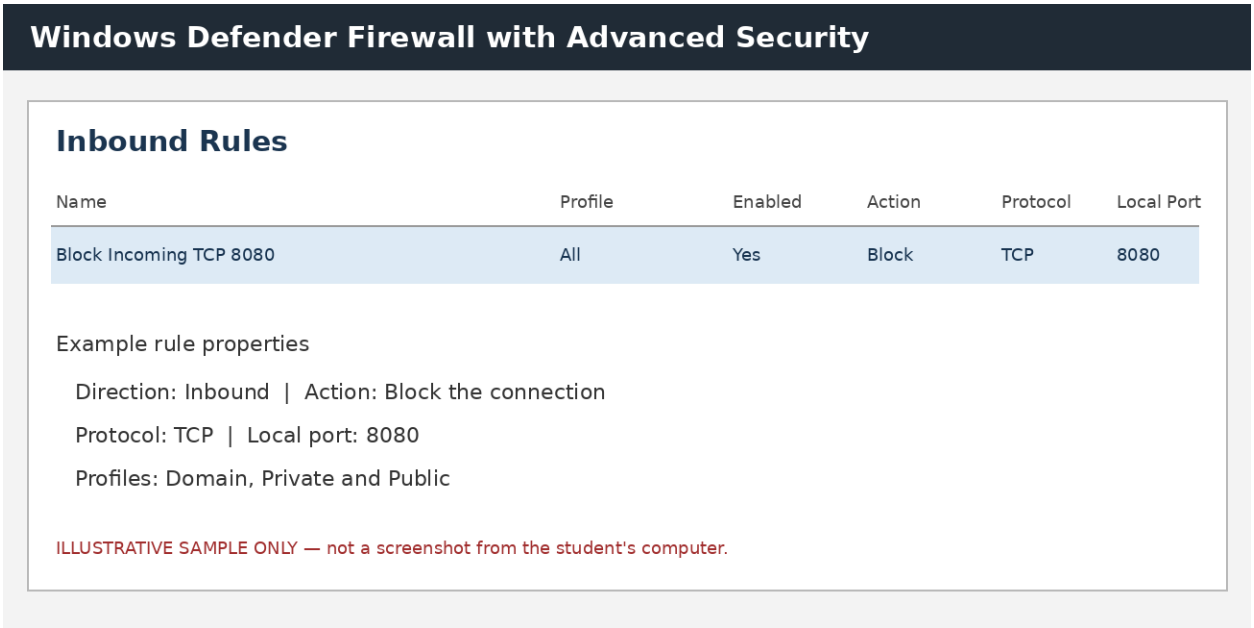
Windows Defender Firewall controls network traffic entering and leaving a computer. An inbound rule can block connection attempts addressed to a particular local port. The steps below create a rule that blocks TCP traffic arriving on local port 8080.

1. Open Start, search for “Windows Defender Firewall with Advanced Security”, and launch it. Administrator permission may be required.
2. Select Inbound Rules in the left pane, then choose New Rule… in the Actions pane.
3. Choose Port as the rule type and click Next.
4. Choose TCP, select Specific local ports, enter 8080, and click Next.
5. Choose Block the connection and click Next.
6. Select Domain, Private and Public so the rule applies to all network profiles, then click Next.
7. Name the rule “Block Incoming TCP 8080”, click Finish, and verify that it appears in Inbound Rules and is enabled.

Expected rule settings:

Setting	Value
Rule name	Block Incoming TCP 8080
Direction	Inbound
Protocol	TCP
Local port	8080
Action	Block the connection
Profiles	Domain, Private and Public
Enabled	Yes

Illustrative example of how the completed rule may appear:



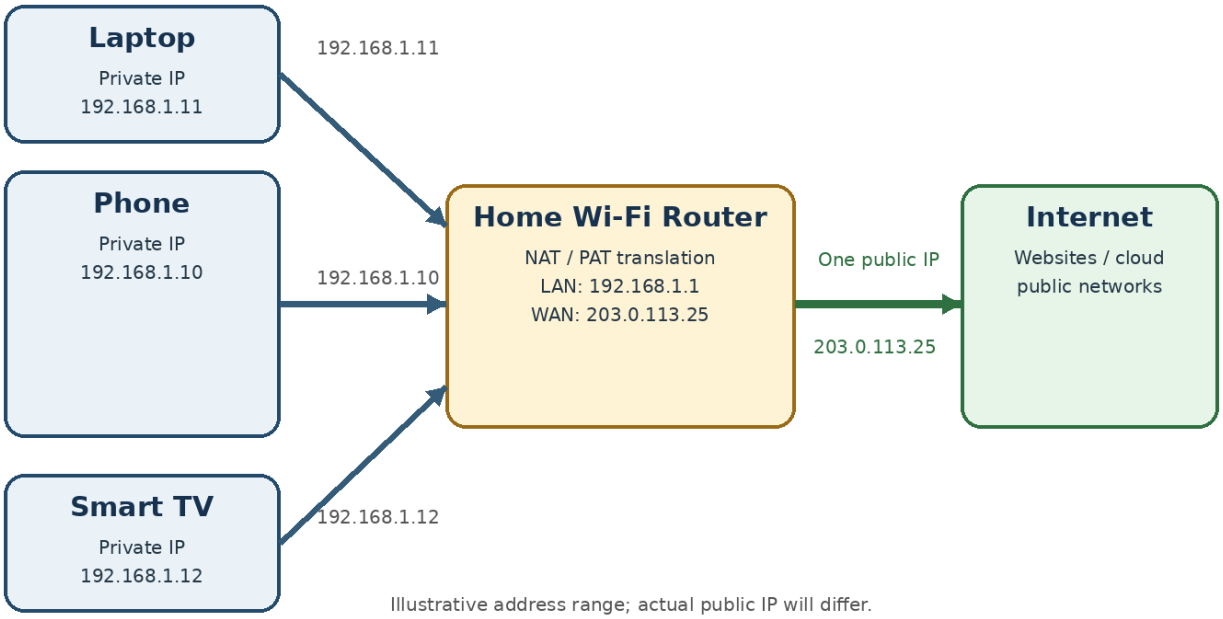
Screenshot requirement: The image above is illustrative only, not a capture from this computer. For submission, take your own screenshot after creating the rule. Show the Inbound Rules list with the rule name, Enabled status and Block action, or capture its Properties window showing TCP and local port 8080.

TASK 2

Simulate NAT (Network Address Translation) by drawing a simple diagram showing how a home Wi-Fi router assigns private IP addresses to a phone, laptop and smart TV, and translates them to a single public IP for Internet access.

Answer

Network Address Translation (NAT) is commonly performed by a home router. The router assigns private addresses to devices on the local network, often using DHCP. When a device accesses the Internet, the router replaces the private source address with the public address assigned to its WAN interface. Port Address Translation (PAT), also called NAT overload, tracks individual connections using port numbers so several devices can share one public IPv4 address.



Example address allocation:

Device / interface	IP address	Purpose
Phone	192.168.1.10	Private LAN address
Laptop	192.168.1.11	Private LAN address
Smart TV	192.168.1.12	Private LAN address
Router LAN	192.168.1.1	Default gateway
Router WAN	203.0.113.25	Example public address toward Internet

When the devices send requests to Internet services, the router records each connection in its NAT/PAT table. Replies return to the router's public address, and the table allows it to forward each reply to the correct private device. The address 203.0.113.25 is from a documentation-only example range; a real router will normally receive a different address from its Internet Service Provider.

TASK 3

Configure port filtering using the Wi-Fi router's admin panel (or a router simulator if the router is not accessible). Block outgoing traffic on port 25 (SMTP) and explain in two sentences how this could help prevent spam from the network.

Answer

Port filtering checks packet headers and permits or denies traffic according to protocol, direction and port. SMTP commonly uses TCP port 25 for server-to-server mail transfer. Blocking outbound TCP port 25 at the router can reduce the ability of infected or misconfigured devices to send email directly to external mail servers.

General configuration procedure (menu names vary by router):

8. Connect to the home Wi-Fi network and sign in to the router administrator interface using the documented gateway address.
9. Open Firewall, Security, Access Control or IP/Port Filtering settings.
10. Create an outbound rule with protocol TCP, destination port 25, action Deny/Drop/Block, and apply it to LAN clients or all devices.
11. Save or apply the setting. If logging is available, enable it and verify that a test connection is blocked.

If using a simulator, create the equivalent outbound TCP destination-port 25 deny rule and record its settings. Do not change router settings unless authorised to administer that network.

Two-sentence explanation: Blocking outbound TCP port 25 can stop compromised computers or malicious programs from sending large volumes of unsolicited email directly to external mail servers. It reduces one common route for spam distribution, although it does not stop all spam because legitimate mail services may use authenticated submission on other ports, such as 587 or 465.

Port-filtering rule summary:

Parameter	Example setting
Direction	Outbound
Protocol	TCP
Destination port	25
Action	Deny / Drop / Block
Scope	All LAN devices, if supported

TASK 4

Given a sample packet-filtering rule set provided by the trainer, identify whether these packets would be allowed or blocked: (1) HTTP from 192.168.1.5 to 172.217.166.78:80, (2) SSH from 10.0.0.10 to 34.201.23.1:22, and (3) DNS from 192.168.1.7 to 8.8.8.8:53.

Answer

Important assumption: The trainer's actual rule set was not included in the task prompt. Therefore, this worked example uses the sample policy below. If the trainer supplies different rules, recalculate the decisions using those rules.

Assumed sample outbound policy (evaluated from top to bottom; first match wins):

Priority	Source	Destination / port	Action
1	Any	Any destination, TCP 22	BLOCK
2	192.168.1.0/24	Any destination, TCP 80	ALLOW
3	192.168.1.0/24	8.8.8.8, UDP/TCP 53	ALLOW
4	Any	Any other traffic	BLOCK

Packet-by-packet decision:

Packet	Source IP	Destination IP:port	Matching rule	Decision
1. HTTP	192.168.1.5	172.217.166.78:80	Rule 2: LAN source, TCP 80	ALLOWED
2. SSH	10.0.0.10	34.201.23.1:22	Rule 1: TCP 22 blocked	BLOCKED
3. DNS	192.168.1.7	8.8.8.8:53	Rule 3: DNS to 8.8.8.8	ALLOWED*

*Packet 3 is allowed if it is a DNS query using UDP or TCP port 53 and the source network matches 192.168.1.0/24. If the trainer's policy does not include this DNS allow rule, or the transport protocol differs, the result may change.

Method: compare the source IP with the permitted source range, identify the destination IP and port, check the transport protocol, and apply the first matching rule. A packet that does not match an allow rule is blocked by the assumed final-deny rule.

TASK 5

Research and list three ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.

Answer

Large online platforms such as Instagram and Zomato operate public-facing applications, APIs, databases and internal services. The following are common defensive practices that a platform of this kind might use; they are general examples and do not claim to disclose either company's private network configuration.

1. Restricting exposed ports and services

Network firewalls can allow public traffic only to required entry points, such as HTTPS on TCP port 443, while blocking direct Internet access to databases, administration interfaces and unused ports. Administrative services can be restricted to approved management networks or secure access gateways.

2. Filtering suspicious traffic and limiting abuse

Firewall rules and upstream filtering systems can drop packets that match known malicious patterns, invalid connection states, spoofed source ranges or unwanted protocols. Rate limits and related traffic controls can reduce abusive connection floods, scanning and automated misuse; specialised DDoS protection may be used for large volumetric attacks.

3. Separating network zones and protecting internal services

Firewalls or cloud network security rules can separate public web servers, application services, databases and management systems into different network zones. For example, a public web tier may contact an application tier on specific ports, while database access is limited to approved application servers rather than the open Internet.

Summary comparison of the three protection methods:

Protection method	Main purpose	Example control
Port/service restriction	Reduce exposed attack surface	Allow HTTPS; deny unused ports
Traffic filtering / abuse control	Reject unwanted or excessive traffic	Packet rules, rate limits, DDoS filtering
Network segmentation	Limit movement between systems	Only application servers can reach database port

Conclusion

Firewalls and packet filtering are important layers of network defence. They help enforce which systems can communicate, which services are reachable and which traffic should be rejected. They work best alongside secure application code, authentication, patching, monitoring, encryption and incident-response procedures.